

HandsOn 4-Template-Driven Forms & Validation

Aim:

To create a Template-Driven Form in Angular using ngModel and ngForm, apply built-in validations, display validation messages, and handle form submission for a Student Enrollment Request.

Procedure

1. Create an Angular project.
2. Generate the EnrollmentForm component.
3. Import FormsModule.
4. Create the enrollment form using ngForm.
5. Bind controls using [(ngModel)].
6. Apply validators (required, minlength, email).
7. Display validation messages.
8. Disable the Submit button when the form is invalid.
9. Display a success message after submission.
10. Reset the form using resetForm().

Codes:

Enrollment-form.ts:

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { FormsModule, NgForm } from '@angular/forms';
@Component({
  selector: 'app-enrollment-form',
  standalone: true,
  imports: [CommonModule, FormsModule],
  templateUrl: './enrollment-form.html',
  styleUrls: ['./enrollment-form.css']
})
export class EnrollmentForm {
  studentName = '';
```

```

studentEmail = "";
courseId: number | null = null;
preferredSemester = "";
agreeToTerms = false;
submitted = false;
onSubmit(form: NgForm) {
  console.log('Form Value:', form.value);
  console.log('Form Valid:', form.valid);
  if (form.valid) {
    this.submitted = true;
  }
}
}
}

```

Enrollment-form.html:

```

<h2>Student Enrollment Request Form</h2>
<form #enrollForm="ngForm" (ngSubmit)="onSubmit(enrollForm)">
  <!-- Student Name -->
  <div>
    <label>Student Name</label><br>
    <input
      type="text"
      name="studentName"
      [(ngModel)]="studentName"
      #nameCtrl="ngModel"
      required
      minlength="3">
    <br>
    <span
      *ngIf="nameCtrl.touched && nameCtrl.errors?.['required']"

```

```
class="error">
  Name is required
</span>
<span
  *ngIf="nameCtrl.touched && nameCtrl.errors?.['minlength']"
  class="error">
    Name must be at least 3 characters
  </span>
</div>
<br>
<!-- Student Email -->
<div>
  <label>Email</label><br>
  <input
    type="email"
    name="studentEmail"
    [(ngModel)]="studentEmail"
    #emailCtrl="ngModel"
    required
  email>
  <br>
  <span
    *ngIf="emailCtrl.touched && emailCtrl.errors?.['required']"
    class="error">
      Email is required
    </span>
  <span>
    *ngIf="emailCtrl.touched && emailCtrl.errors?.['email']"
    class="error">
```

Enter a valid email

</div>

<!-- Course ID -->

<div>

<label>Course ID</label>

<input

type="number"

name="courseId"

[(ngModel)]="courseId"

#courseCtrl="ngModel"

required>

*ngIf="courseCtrl.touched && courseCtrl.errors?.['required']"

class="error">

Course ID is required

</div>

<!-- Preferred Semester -->

<div>

<label>Preferred Semester</label>

<select

name="preferredSemester"

[(ngModel)]="preferredSemester"

required>

<option value="">Select Semester</option>

```
<option value="Odd">Odd</option>
<option value="Even">Even</option>
</select>
</div>
<br>
<!-- Agree To Terms -->
<div>
  <input
    type="checkbox"
    name="agreeToTerms"
    [(ngModel)]="agreeToTerms"
    #termsCtrl="ngModel"
    required>
  <label>I Agree to Terms & Conditions</label>
  <br>
  <span
    *ngIf="termsCtrl.touched && termsCtrl.errors?.['required']"
    class="error">
    You must accept the terms.
  </span>
</div>
<br>
<!-- Buttons -->
<button
  type="submit"
  [disabled]="enrollForm.invalid">
  Submit
</button>
&nbsp;
```

```
<button
  type="button"
  (click)="enrollForm.resetForm()">
  Reset
</button>
</form>
<br>
<!-- Success Message -->
<div
  *ngIf="submitted"
  class="success">
  Enrollment request submitted successfully!
</div>
```

Enrollment-form.css:

```
/* Form Container */
form {
  width: 450px;
  margin: 20px auto;
  padding: 20px;
  border: 1px solid #cccccc;
  border-radius: 8px;
  background-color: #f9f9f9;
  box-shadow: 0px 2px 6px rgba(0,0,0,0.2);
}

/* Heading */
h2 {
  text-align: center;
  color: #1976d2;
}
```

```
/* Labels */
label {
    font-weight: bold;
}

/* Inputs */
input,
select {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
    margin-bottom: 10px;
    border-radius: 5px;
    border: 1px solid #cccccc;
    box-sizing: border-box;
}

/* Validation Colors */
.ng-invalid.ng-touched {
    border: 2px solid red;
}

.ng-valid.ng-touched {
    border: 2px solid green;
}

/* Buttons */
button {
    padding: 10px 18px;
    margin-right: 10px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
```

```
    background-color: #1976d2;
    color: white;
}
button:hover {
    background-color: #1259a7;
}
button:disabled {
    background-color: gray;
    cursor: not-allowed;
}
/* Error Messages */
.error {
    color: red;
    font-size: 13px;
}
/* Success Message */
.success {
    width: 450px;
    margin: 20px auto;
    padding: 12px;
    text-align: center;
    background-color: #d4edda;
    color: #155724;
    border: 1px solid #c3e6cb;
    border-radius: 5px;
    font-weight: bold;
}
```

App.route.ts:

```
import { Routes } from '@angular/router';
```



```
import { EnrollmentForm } from './pages/enrollment-form/enrollment-form';

export const routes: Routes = [

  {
    path: "",
    redirectTo: 'enroll',
    pathMatch: 'full'
  },
  {
    path: 'enroll',
    component: EnrollmentForm
  }
];
```

App.ts:

```
import { Component } from '@angular/core';
import { RouterOutlet } from '@angular/router';

@Component({
  selector: 'app-root',
  standalone: true,
  imports: [RouterOutlet],
  templateUrl: './app.html',
  styleUrls: ['./app.css']
})

export class App {
}
```

App.html:

```
<h1 style="text-align:center">
  Student Course Portal
</h1>
```

<hr>

<router-outlet></router-outlet>

App.config.ts:

<h1 style="text-align:center">

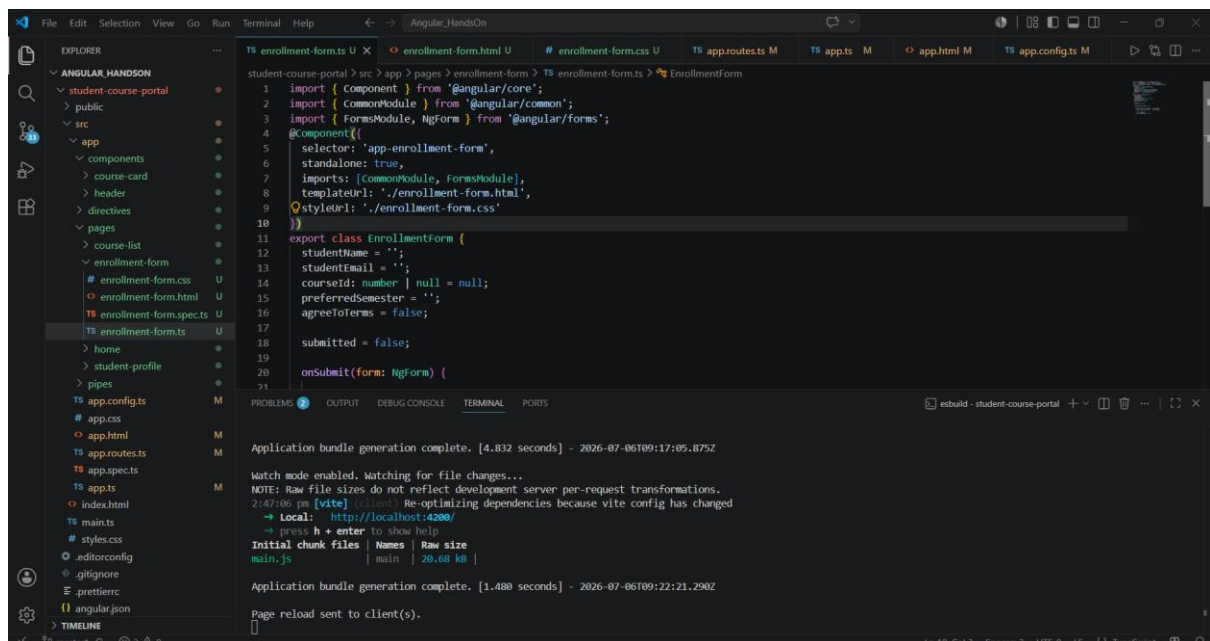
Student Course Portal

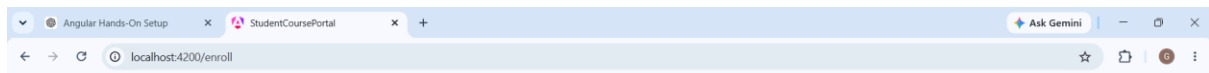
</h1>

<hr>

<router-outlet></router-outlet>

Outputs:





Student Course Portal

Student Enrollment Request Form

Student Name

GAYATHRI VANGA

Email

vangagayathri@gmail.com

Course ID

3

Preferred Semester

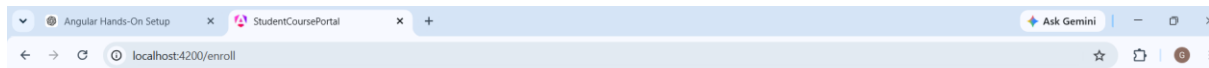
Even

☒

I Agree to Terms & Conditions

Submit

Reset



Student Course Portal

Student Enrollment Request Form

Student Name

GAYATHRI VANGA

Email

vangagayathri@gmail.com

Course ID

3

Preferred Semester

Even

☒

I Agree to Terms & Conditions

Submit

Reset

Enrollment request submitted successfully!

Result

Successfully created a Template-Driven Student Enrollment Form with validation, error messages, successful submission handling, and reset functionality using Angular 20.